

Zhuoran Li

+86 151-8984-3657 | airplane@sjtu.edu.cn | [airplane.github.io](https://github.com/airplane)

RESEARCH INTERESTS

Reliable vision-language-action models; embodied AI and robot learning; failure diagnosis and recovery; memory-augmented agents; long-horizon manipulation; multimodal and learning systems.

EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China Sep. 2023 – Jun. 2027
(Expected)

B.Eng. in Software Engineering, School of Computer Science

GPA: **92.2/100**; Rank: **3/95**. Undergraduate researcher at the Institute of Parallel and Distributed Systems (IPADS), advised by Prof. Jinyu Gu.

SELECTED PUBLICATIONS

RoboHarness: A Memory-Augmented Policy Harness for Vision-Language-Action Model Robustness via In-Context Adaptation

Zhuoran Li, Zhiyang Li, Kaijun Zhou, and Jinyu Gu.

IROS, 2026. [Project] [Paper] [Code]

GKD-Recruiter: Jointly Modeling Social and Task Heterogeneity for Spatial Crowdsourcing via Graph Knowledge Distillation

Yucen Gao, Zhemeng Yu, Zhuoran Li, Jianxiong Guo, and Xiaofeng Gao.

ICML, 2026. [Project] [Paper] [Code]

DRNCS: Dual-Level Route Generation Model Based on Node Contraction and Shortcuts

Zhuoran Li, Yucen Gao, Yu Yin, Xinle Li, Hui Gao, Xiaofeng Gao, and Guihai Chen.

ECML PKDD, 2025. [Paper] [Code]

Algorithms for Shortest Path Tour Problem

Yucen Gao, Zhuoran Li, Jingyu He, Jun Fang, Hui Gao, Xiaofeng Gao, and Guihai Chen.

Theoretical Computer Science, 1054:115428, 2025. [Paper]

RESEARCH EXPERIENCE

RoboHarness: Memory-Augmented Policy Harness for VLA Robustness Dec. 2025 – Jun. 2026

Co-first Author (first-listed) | Accepted at IROS 2026

- Designed an inference-time reliability harness around frozen VLA policies, combining contrastive Dual-Memory retrieval, attribution-driven vision-language orchestration, MCP-based tool interventions, and offline memory consolidation.
- Led framework design, core implementation, experiments, and manuscript preparation; evaluated π_0 , $\pi_{0.5}$, and SmoVLA on LIBERO-PRO and LIBERO-RoboHarness without parameter fine-tuning, yielding a 56.6-percentage-point average absolute success-rate gain and up to 89.1 percentage points on long-horizon tasks.

ManiLadder: Multi-Level Benchmark for Manipulation Intelligence Sep. 2025 – Present

Core Developer and Manuscript Contributor | Manuscript in preparation

- Built a modular simulation and data pipeline spanning task construction, demonstration collection, policy training, and automated evaluation; the benchmark contains **114 tasks**, **5,700 human demonstrations**, and four difficulty levels.
- Contributed to the benchmark’s simulation environments, reproducible evaluation workflow, and manuscript, supporting unified comparison of imitation-learning and VLA policies across task levels.

RESEARCH EXPERIENCE (CONTINUED)

GKD-Recruiter: Graph Knowledge Distillation for Spatial Crowdsourcing Jan. 2025 – Jun. 2026

Researcher | ICML 2026 regular paper accepted

- Studied spatial task allocation with heterogeneous task requirements and worker attributes through task-worker graph representations and graph knowledge distillation.
- Contributed to algorithm modeling, experimental analysis, and manuscript preparation, with an emphasis on learning task-worker matching relations at scale.

DRNCS: Dual-Level Route Generation via Node Contraction and Shortcuts Jun. 2024 – Oct. 2024

Sole First Author | Published at ECML PKDD 2025

- Proposed shortcut-edge differential contraction and dynamic probabilistic shortcut injection to compress dense road-network search spaces; achieved **0.3–0.8 ms/trip** latency and **3,356 trips/s** throughput.
- Independently carried out method design, system implementation, experimental validation, and manuscript writing from problem formulation to publication.

Algorithms for the Shortest Path Tour Problem (SPTP) Nov. 2024 – Mar. 2026

Algorithm Researcher and Co-Author | Published in Theoretical Computer Science

- Contributed to problem formalization, Stage and Global Dijkstra algorithms, and complexity analysis, achieving $O(\ell(n + m) \log n)$ time and $O(\ell n)$ space for exact SPTP search.
- Evaluated the algorithms on million-node road networks and obtained up to a **50× speedup** over baseline methods.

RESEARCH APPOINTMENTS AND INTERNATIONAL EXPERIENCE

Institute of Parallel and Distributed Systems (IPADS), SJTU Jun. 2025 – Present

Undergraduate Research Intern; Advisor: Prof. Jinyu Gu

LinS Lab, National University of Singapore (NUS) Summer 2025

Summer Research Intern

Imperial College London Winter 2025

Winter School in Data Science and Artificial Intelligence

ONGOING RESEARCH

- Contributing to the system-software taxonomy, literature review, and manuscript preparation for *Beyond the Model: A Survey of System Software for Embodied Agents* (2026–Present).
- Helped organize the literature around planning, skill and tool management, execution control, memory and world state, verification and recovery, adaptation, and multi-agent coordination.

SELECTED RESEARCH GRANTS AND HONORS

NSFC Basic Research Program for Undergraduate Students (2026)

Nationally Selected Student PI | *Research on OpenHarmony Embodied Agent Systems* | RMB 100,000

Chun-Tsung Foundation Award (2024) | **Student Principal Investigator**

Zhiyuan Honor Scholarship (2023–2025); Academic Scholarship (2024–2025)

First Prize, Shanghai Undergraduate Mathematics Competition (Non-Math Category A, 2024)

Jiangsu Province Excellent Student Award (2023)

TECHNICAL SKILLS

Programming C++, Python, Java, JavaScript/TypeScript

AI and Robotics PyTorch; computer vision; reinforcement learning; graph learning; imitation learning; VLA and multimodal evaluation

Systems and Tools Linux, Git, LaTeX, React, Spring Boot, Nginx, Redis

English CET-4: 624; CET-6: 605